RAFAEL SAN VIDAL

Senior Software Engineer | Clinical Laboratory Systems & Hospital Integration | .NET · HL7 · Healthcare Interoperability

25+ YEARS PROFESSIONAL EXPERIENCE | 20+ YEARS HEALTHCARE EXPERTISE

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PROFESSIONAL PROFILE

Senior Software Engineer specialized in **clinical laboratory systems and hospital integration**, with 20+ years at Becton Dickinson building software deployed in **40+ hospitals**. I design and build platforms that connect laboratory instruments, clinical analyzers, middleware, LIS and HIS (HL7, ASTM, SOAP, REST, TCP/IP sockets, RFID), with experience integrating laboratory ecosystems from Siemens Healthineers, Werfen (Modulab), Roche Diagnostics and others.

My focus is patient safety and data integrity across the pre-analytical, analytical and post-analytical workflow: positive patient and sample identification, end-to-end sample traceability, tamper-evident audit trails and risk-based engineering (ISO 14971). 25+ years of professional experience, full-stack across .NET and Java.

> **Current Focus:** Rebuilding BD's hospital laboratory software on .NET 9 with Clean Architecture and DDD: 83 documented architecture decisions (ADRs) and ~11,500 automated tests. AI-assisted development (Claude Code, Cursor) used under documented architecture rules and automated testing, to accelerate delivery without compromising quality.

HEALTHCARE EXPERTISE

Clinical Laboratory Workflow

- Pre-analytical: phlebotomy rooms, lab request forms, sample labelling
- Positive patient & sample identification
- Sample tracking with RFID, from collection to laboratory
- Globally unique sample numbering, also offline
- Analytical & post-analytical: instrument integration, result management

Interoperability

- Instruments, analyzers & middleware to LIS and HIS
- Siemens Healthineers, Werfen (Modulab), Roche Diagnostics
- HL7 v2 over TCP/IP sockets, HL7-FHIR, ASTM
- SOAP, REST, direct database connectors
- Mirth Connect integration engine

Regulatory & Patient Safety

- ISO 14971 risk management
- GDPR / LOPDGDD for health data, ENS, ISO 27001
- HMAC-signed audit trail, AES-GCM column encryption
- Patient data isolated in dedicated read-only services

AI-ASSISTED DEVELOPMENT TOOLS (SINCE 2024)

Cursor Pro	AI-powered code editor · Integrated
Claude Code	CLI-based AI assistant · Integrated
NotebookLM	AI research tool · Research

PROFESSIONAL EXPERIENCE

Becton Dickinson and Company — Senior Software Developer | Healthcare Solutions — 2005 – Present (20+ years)

Key Responsibilities

- Hospital laboratory software, EMEA region (40+ hospitals)
- Full-stack development, architecture design, UI/UX, security
- CI/CD, DevOps, database management
- Team mentoring and code review

Healthcare Domain (20+ years)

- Phlebotomy room & lab request workflows
- HIS/LIS integration: HL7/HL7-FHIR, ASTM, Mirth Connect
- RFID sample tracking in hospitals
- Patient & sample identification, traceability

Quality & Security

- Fortify OnDemand (SAST), SonarQube
- OWASP Top 10, vulnerability assessment
- GDPR / ENS compliance, ISO 14971 risk analysis
- Penetration testing, code coverage

Multi-Database Expertise

- MySQL, MariaDB, SQL Server, Oracle, DB2
- Flyway, Entity Framework migrations
- Cross-database compatibility

Technologies: .NET 9, C# 13, Java 21, Spring Boot, Spring Data JPA, Spring Security, React, Angular, TypeScript, Blazor, Thymeleaf, Bootstrap, Tailwind, SQL Server, MySQL, Oracle, IBM DB2, MariaDB, Azure DevOps, Docker, Git, Jira, SignalR, WebSockets, Socket.io, WPF, WinUI, Uno Platform, RCP / RAP

Omnibona S.L — Software Developer (Part-time) | Multi-platform Health Application 2014 – 2015

Design and development of multi-platform application about healthy habits. Nutrition and exercises in game mode for startup company.

Technologies: ASP.NET MVC, Razor, Entity Framework, LINQ, HTML5, CSS3, Bootstrap, JavaScript, jQuery, Microsoft Azure

Oesia — Software Developer 2003 – 2005

Healthcare, Telecom, Public Services. Clients: Telefónica, HP, Siemens

Technologies: .NET, C#, SQL Server, Oracle

Easynet España S.A. — Support Technical 2003

Technical support and customer service

L'Oréal Spain — SAP Key Business | Business Analyst 1998 – 2003 (5 years)

SAP R/3 Project Leadership

- Business Key for SAP R/3 (FI, CO, MM)
- Project planning, design, implementation
- ABAP IV programming, reports development
- Business requirements, stakeholder mgmt

Historical Projects

- Y2K and Euro currency conversion
- Accounts and Purchasing systems
- AS/400 and IBM AIX integrations

Technologies: SAP R/3, ABAP IV, C++, RPG, 4GL, AS/400, IBM AIX, Oracle, CA Ingres

Spanish Ministry of Foreign Affairs — Software Developer | Diplomatic Passports Management 1998

Design and development of application to manage diplomatic passports

Technologies: 4GL, CA Ingres

COMPLETE TECHNOLOGY STACK

Backend / Languages	.NET 9 / C# 13, Java 21, Spring Boot 3.x, ASP.NET Core 9, Node.js / Express, Python / Django, PHP / Laravel, Erlang / Haskell
Frontend	React, Next.js, Angular, TypeScript, Blazor WebAssembly, WPF / WinForms, Uno Platform / WinUI, HTML5 / CSS3, Tailwind / Bootstrap, Vue.js
Databases	SQL Server, MySQL / MariaDB, PostgreSQL, Oracle, IBM DB2, SQLite, MongoDB, Redis, Cassandra, InterSystems Caché, Informix Dynamic Server, CA Ingres
Cloud & DevOps	Azure / AWS, Azure DevOps, Docker, Kubernetes, Git / GitHub, GitHub Actions, Jenkins, Subversion / CVS, CI/CD Pipelines
Testing	JUnit / Mockito, xUnit / nUnit / Moq, Selenium, Playwright, Jest / Vitest, Jasmine / Karma, TDD / BDD
Spring Ecosystem	Spring Data JPA, Spring Security + OAuth2, Spring MVC / WebFlux, Spring AOP, Spring Validation, Spring Boot DevTools, Thymeleaf + Security Extensions, Flyway, Spring Cloud
Java Ecosystem	Hibernate / JPA / Criteria API, Maven / Gradle, Apache CXF / Jersey, HikariCP
.NET Ecosystem	EF Core / Dapper / LINQ, Minimal APIs, Swagger / OpenAPI, MediatR, NLog / Serilog, AutoMapper, Polly.NET, SignalR, Hangfire, FluentValidation, MailKit, QuestPDF
Architecture	Clean Architecture, DDD, CQRS, Microservices, Event-Driven, SOA
Methodologies	Scrum / Kanban, XP, TDD / BDD, SOLID Principles, Clean Code, Design Patterns
Quality & Security	Fortify OnDemand (SAST), SonarQube, OWASP Top 10, GDPR / ENS, Penetration Testing, Code Coverage Analysis
Mobile Development	Kotlin, Android Studio
Historical Languages	ABAP IV (SAP), RPG (AS/400), 4GL (Ingres), COBOL, Pascal, C++

Enterprise Systems	SAP R/3 (FI, CO, MM), ABAP IV Programming, Business Analysis, AS/400 Systems, IBM AIX
Big Data & Analytics	IBM Cognos, Talend ETL, Elasticsearch, Prometheus, Grafana
Microservices & Messaging	Apache Kafka, RabbitMQ, Spring Cloud (Eureka, Hystrix), Event-Driven Architecture
Specialized Technologies	Blockchain & Web3 (Solidity, Ethereum), Game Development (Unity 3D, C#), IoT (Arduino, Raspberry Pi), Computer Vision (Infaimon CVB), Document Recognition & Processing (Abbyy Technologies), CMS Development (WordPress, Joomla)
Application Servers	Apache Tomcat, Microsoft IIS, Oracle WebLogic, WildFly, IBM WebSphere
Project Management	Jira, Confluence
AI Tools	Cursor, Claude Code, Azure OpenAI Services, Semantic Kernel, NotebookLM
Development Tools	Visual Studio Professional, Visual Studio Code, Cursor, Eclipse, IBM Rational Suite, Postman, Insomnia

METHODOLOGIES & SOFTWARE ENGINEERING PRINCIPLES

Agile Methodologies

- Scrum (Sprint Planning, Daily Standups, Retrospectives)
- Kanban (Continuous Flow, WIP limits)
- XP (Extreme Programming)
- Lean Software Development

Architecture Patterns

- Clean Architecture
- Domain-Driven Design (DDD)
- CQRS (Command Query Responsibility Segregation)
- Repository Pattern
- Microservices Architecture
- Event-Driven Architecture
- SOA (Service Oriented Architecture)

SOLID Principles

- Single Responsibility Principle
- Open/Closed Principle
- Liskov Substitution Principle
- Interface Segregation Principle
- Dependency Inversion Principle

Testing Methodologies

- TDD (Test-Driven Development)
- BDD (Behavior-Driven Development)
- ATDD (Acceptance Test-Driven Development)
- Unit Testing / Integration Testing

Design Patterns (GoF)

- Creational: Factory, Builder, Singleton
- Structural: Adapter, Decorator, Facade
- Behavioral: Observer, Strategy, Command

Code Quality

- Clean Code Principles
- Code Review Practices
- Refactoring Techniques
- Technical Debt Management
- Documentation Standards

FEATURED PROFESSIONAL PROJECTS (HEALTHCARE)

BD Healthcare Platform — BDWiderlab · BDSupport · BDForms

Modular monolith rebuilding BD's hospital software on .NET 9, since 2025

- Three products on one shared framework: medical backend for lab requests (BDWiderlab), support & equipment inventory (BDSupport), and digital lab request forms with PDF417 barcodes replacing OMR paper forms (BDForms)
- Patient safety by design: globally unique sample numbers even on offline stations (append-only range ledger), positive patient identification, ISO 14971 risk analysis
- HIS/LIS integration through isolated read-only services (BDPatient, BDLis) with typed SDKs: patient data kept separate, the product degrades gracefully if a hospital system is down
- Tamper-evident audit trail: every audit record HMAC-signed with a key kept outside the database (GDPR art. 32, ENS, ISO 27001)
- Clean Architecture + DDD, CQRS with MediatR, Specifications, Outbox pattern, optimistic concurrency; REST endpoints as Minimal APIs documented with Swagger/OpenAPI
- Security: JWT with refresh tokens + MFA (TOTP), RBAC with row-level authorization and token revocation, AES-GCM column encryption, signed audit trail
- Five database providers from one codebase (SQLite, SQL Server, MySQL, Oracle, DB2), with per-provider EF Core migrations
- Web: Next.js 15 / React 19 / TypeScript with a shared component library; BDForms on Uno Platform (WinUI 3, WebAssembly, Android) with TWAIN scanning and QuestPDF printing
- ~11,500 unit/integration tests (xUnit, Moq) + Vitest frontend tests + Playwright E2E; 83 ADRs; Azure DevOps CI/CD
- Built with an AI-assisted workflow (Claude Code agent teams guided by project rules and ADRs)

Tech stack: .NET 9, C# 12, EF Core 9, Minimal APIs, Swagger/OpenAPI, MediatR, FluentValidation, Serilog, Polly, QuestPDF, Next.js 15, React 19, TypeScript, Tailwind, Uno Platform, SQL Server, Oracle, DB2, MySQL, SQLite, xUnit, Moq, Vitest, Playwright, Azure DevOps

BD RFID System — Healthcare RFID Management

Full-stack enterprise platform for hospital sample tracking

- Tag Manage: RFID tag management interface
- Tag Backend: REST API with multi-database support
- SSTSDaemon: Background service for sample tracking
- Tag Shipping: PHP/Laravel shipping module
- Clean Architecture with Domain-Driven Design
- Real-time dashboard with WebSockets (Stomp.js)
- JWT authentication & session management
- WinUI 3 desktop application
- RFID reader integration (CAEN, Impinj)
- Distributed caching & high availability
- In production across 40+ hospitals

Tech stack: Spring Boot, Thymeleaf, Angular, TypeScript, .NET, C#, PHP, Laravel, Node.js, Express, Oracle, SQL Server, IBM DB2, MySQL, WebSockets, RFID, DDD, REST API, JWT, Swagger

BD WiderLab — Phlebotomy Room Management

Integral management of extraction rooms with HL7 messaging broker

- End-to-end workflow for phlebotomy room operations: patient identification, sample collection and labelling
- HL7 messaging broker for interoperability with hospital systems (HIS/LIS): patient demographics, lab orders and samples
- Integration with diagnostic devices
- In production across 40+ hospitals

Tech stack: Java, Eclipse, HL7

ScanBatch OCR — Document Processing System

Batch scanning with OCR and barcode recognition

- Abbyy SDK integration for high-quality OCR
- Barcode recognition library for automated processing
- Scanned lab request forms linked to the right patient through pluggable connectors to each hospital's systems (HL7 over sockets, SOAP, REST, SQL)
- TWAIN scanner driver support for hardware integration
- Windows Service daemon for background processing
- In production across 40+ hospitals

Tech stack: WPF, C#, Abbyy OCR, TWAIN, WIA, Windows Services

LEARNING & EXPERIMENTAL PROJECTS

Azure AI Samples & Training

Microsoft official AI samples and courses

- Microsoft official ai-samples repository (40+ examples)
- Semantic Kernel tutorials (Build 2024): Hello SK, Chat History, Plugins, Logging
- Quickstarts: Text Summary, Chat App, Function Calling
- Local models: Phi, LLaMA examples
- AI courses 2024: Prompt Engineering, AI Development, Claude Code integration

Tech stack: C#, Azure OpenAI, Semantic Kernel, Microsoft.Extensions.AI, LLM Evaluation, OpenAI API

Cloud Architecture

Azure + AWS deployment, DDD, TDD, IA integration

- Azure deployment with .NET 8
- Azure Functions serverless
- AWS deployment with Spring Boot
- Cloud architecture patterns and best practices

Tech stack: Azure, AWS, .NET 8, Spring Boot, DDD, TDD, Azure Functions, AWS Elastic Beanstalk

React Full Stack

Complete React + Node.js + MongoDB

- Full-stack web application development
- State management with Redux
- TypeScript integration

Tech stack: React, TypeScript, Node.js, Express, MongoDB, Redux

Android Development with Kotlin

Native Android app development course

- Complete Kotlin course with Android app structure
- Firebase authentication with Google OAuth
- Android UI/UX with ConstraintLayout
- Testing with JUnit and Espresso

Tech stack: Kotlin, Android Studio, Gradle, Firebase Auth, JUnit, Espresso

Microservices with Kubernetes & Azure

Real deployed microservices architecture

- 3 microservices deployed to Azure Kubernetes Service
- Services: Gateway, Library (MongoDB), Security (SQL Server)
- Docker images published to Docker Hub
- Kubernetes manifests (deploy.yml, service.yml)
- JWT authentication and load balancing
- GitHub: github.com/rsvidal/microservices-example1

Tech stack: Kubernetes, Azure AKS, Docker, .NET Core, MongoDB, SQL Server, JWT

Cloud Functions

Serverless Computing

- Azure Functions development
- AWS Lambda implementation
- Serverless architecture patterns

Tech stack: .NET 9, Azure Functions, AWS Lambda, Serverless

Microservices Netflix

Spring Cloud Microservices implementation

- Netflix stack implementation
- Service discovery with Eureka
- Circuit breaker pattern with Hystrix

Tech stack: Spring Boot, Eureka, Hystrix, Docker, Kubernetes

IoT & Embedded Systems Suite

Arduino, Ethernet Shield, Bluetooth, Web Services, Mobile Integration

- IoT system with Arduino and Ethernet Shield
- Web Services for remote communication
- CoAP and REST protocols for IoT
- Bluetooth integration for mobile devices
- Xamarin mobile app for remote control

Tech stack: Arduino, Netduino, C++, Ethernet Shield, Bluetooth HC-05, CoAP Protocol, aREST API, Xamarin, Android, Web Services

Blockchain/Web3 Development Suite

Solidity, Web3.js, Truffle, Ethereum, Smart Contracts

- Token ICO (Deka Token) with complete functionality
- Smart contracts for token management
- Ethereum testnet integration
- Web3 frontend for contract interaction
- Wallet integration system

Tech stack: Solidity, Web3.js, Truffle, Ethereum, Smart Contracts, JavaScript

Unity Game Development

2D game development with Unity 2020.3

- Complete 2D game with Unity
- Interactive text adventure game
- Dialogue and narrative system
- Game mechanics implementation
- Custom assets and sprites

Tech stack: Unity 2020.3, C#, Game Development, 2D Graphics

SOFT SKILLS

- Team Leadership & Mentoring Junior Developers
- Code Review & Quality Assurance
- Technical Documentation
- Stakeholder Communication
- Problem Solving & Analytical Thinking
- Agile/Scrum Ceremonies
- Project Management
- Business Requirements Analysis
- Continuous Learning & Adaptability

TRAINING & CONTINUOUS EDUCATION (80+ COURSES SINCE 1998)

2024-2025	*AI-Assisted Development Era:* Comprehensive Training in Cloud Architecture, DDD, TDD and AI · AI-Assisted Development · Introduction to AI and Prompt Engineering · Claude Code CLI · NotebookLM · .NET 9 & C# 13 · Blazor Development · Dapper ORM · DDD Microsoft Microservices · Real-World Sales System (C# + Angular + Xamarin) · React & TypeScript · Uno Platform · Azure DevOps Complete · Jira Advanced
2023	Ionic (App) and .NET Core (Backend) (Udemy) · Go / Golang from the Ground Up (Udemy) · Big Data Fundamentals (Udemy) · Dart & Flutter (Udemy) · Advanced Vue.js (Udemy)
2022	React + Redux (Udemy, Jul 2022) · .NET MAUI with Visual Studio 2022 (Udemy, Jul 2022) · DevOps, Docker, Kubernetes, Microservices and Azure .NET 6 (Udemy, Jul 2022) · Azure DevOps (Udemy, Jun 2022) · AWS (Udemy, Jun 2022)
2021	Go / Golang (Udemy, Aug 2021) · TDD - Test-Driven Development with Java and JUnit (Udemy, Dec 2021) · Design Patterns and SOLID Principles (Udemy, May 2021) · Refactoring Java Code (Udemy, Mar 2021)
2020	DevOps: The Martial Arts of Software (Udemy, Jun 2020)
2019	Java 8 (Functional Programming and Lambdas) (Udemy, Sep 2019) · Microservices with Spring Boot / Netflix Eureka (Udemy, Sep 2019) · Python Programming for Machine Learning (Udemy, Sep 2019) · Python 3 Course (Udemy, Aug 2019) · Video Game Development with Unity (Udemy, Jun 2019) · My First Video Game with Unity (Udemy, Jun 2019) · RESTful API with Laravel (Udemy, May 2019) · Mastering Laravel: From Beginner to Expert (Udemy, May 2019) · Android with Kotlin + Firebase + Google (Udemy, Apr 2019) · Machine Learning (Udemy, Mar 2019) · Django Course (Udemy, Jan 2019)
2018	Implementing Solutions on Microsoft Azure (Udemy, Nov 2018) · Python + Django (Udemy, Jul 2018) · dApp Development on the Ethereum Blockchain (Udemy, Jun 2018) · Introduction to Blockchain, Cryptocurrencies and Ethereum (Udemy, Jun 2018) · Xamarin Forms and Azure (Udemy, Mar 2018) · Advanced Angular 6 (Udemy, Jan 2018) · Advanced Docker Course (Udemy, Jan 2018) · Spring Framework 5 (Udemy, Jan 2018) · Java Data Persistence: Hibernate + Spring JPA (Udemy, Jan 2018)
2017	ASP.NET Core MVC (Udemy, Nov 2017) · Introduction to Haskell (Udemy, Sep 2017) · Xamarin: Build Android and iOS Apps (Udemy, May 2017) · WCF Services (Udemy, Apr 2017)
2016	Git + GitHub (Udemy, Mar 2016) · Web Development with JavaScript, Angular, Node and MongoDB (Udemy, Apr 2016)
2015	SEO for WordPress (Udemy, Feb 2015) · Succeed with SEO (Udemy, Feb 2015)
2014	Community Manager, Social Media (HP Training, May 2014)
2013	Marketing 2.0 (Grupo Fides21, Mar 2013) · Team Building: Leadership Management (Fides21, Dec 2013)
2006	IBM Rational V7: ClearCase, SOA, Business Modeler (Sep 2006) · IBM DB2 (Sep 2006) · Infaimon Sherlock: Machine Vision (May 2006) · Infaimon CVB: Machine Vision (May 2006)
2005	Project Management (Oesia, 160h, Jan 2005)
2004	Visual C++ Programming (Oesia, 40h, May 2004)
2003	SQL Server Administration (Sunion Gesfor, 40h, Nov 2003) · Java Programming (Oesia, 40h, Oct 2003) · Oracle 8i Optimization (MR Informática, 80h, Jun 2003) · PL/SQL (Dayfisa, 40h, Apr 2003) · XML (IBM Learning, 30h, May

	2003) · Oracle Administration (Dayfisa, 40h, May 2003) · Macromedia Flash (IBM Learning, 30h, Mar 2003) · Software Engineering (Sunion Gesfor, Jan-Mar 2003)
2002	SAP MM: Invoice Verification (SAP Spain, 25h, Oct 2002) · Advanced Java Programming (IBM Learning, 25h, Apr 2002) · AIX Administration (IBM Learning, 30h, Apr 2002) · Advanced RPG IV Programming (AS/400) (IBM Learning, 50h, Feb 2002)
2001	Java Programming (Gesein, 100h, Feb-Apr 2001)
1999	COBOL – CICS – DB2 (Cap Gemini, 160h, 1999)
1998	Unix / Oracle Systems Technician (Sunion Gesfor, 350h, 1998) · Computer Technician (Academia Teide, 300h, 1998)

Total courses completed: 80+ | Platforms: Udemy, IBM Learning, Oesia, Cap Gemini, Academia Teide, HP Training, Fides21 · Active learner since 1998 (25+ years of continuous training)

LANGUAGES

Spanish	Native
English	B1

ADDITIONAL INFORMATION

- **Portfolio:** https://rsvidal.github.io - Terminal-style interactive portfolio
- **Philosophy:** 25+ years applying software engineering best practices. Expert in modern development methodologies and architectural patterns. Strong advocate for clean code and quality engineering. Pioneer in AI-assisted development (Cursor, Claude Code)
- **Healthcare Domain:** 20+ years in clinical laboratory software at Becton Dickinson: phlebotomy and pre-analytical workflows, HIS/LIS integration (HL7/HL7-FHIR, ASTM, Mirth Connect), RFID sample tracking, patient and sample identification, GDPR/ENS and ISO 14971
- **Work Arrangements:** Available for new opportunities and collaborations. Open to remote/hybrid work arrangements in EU